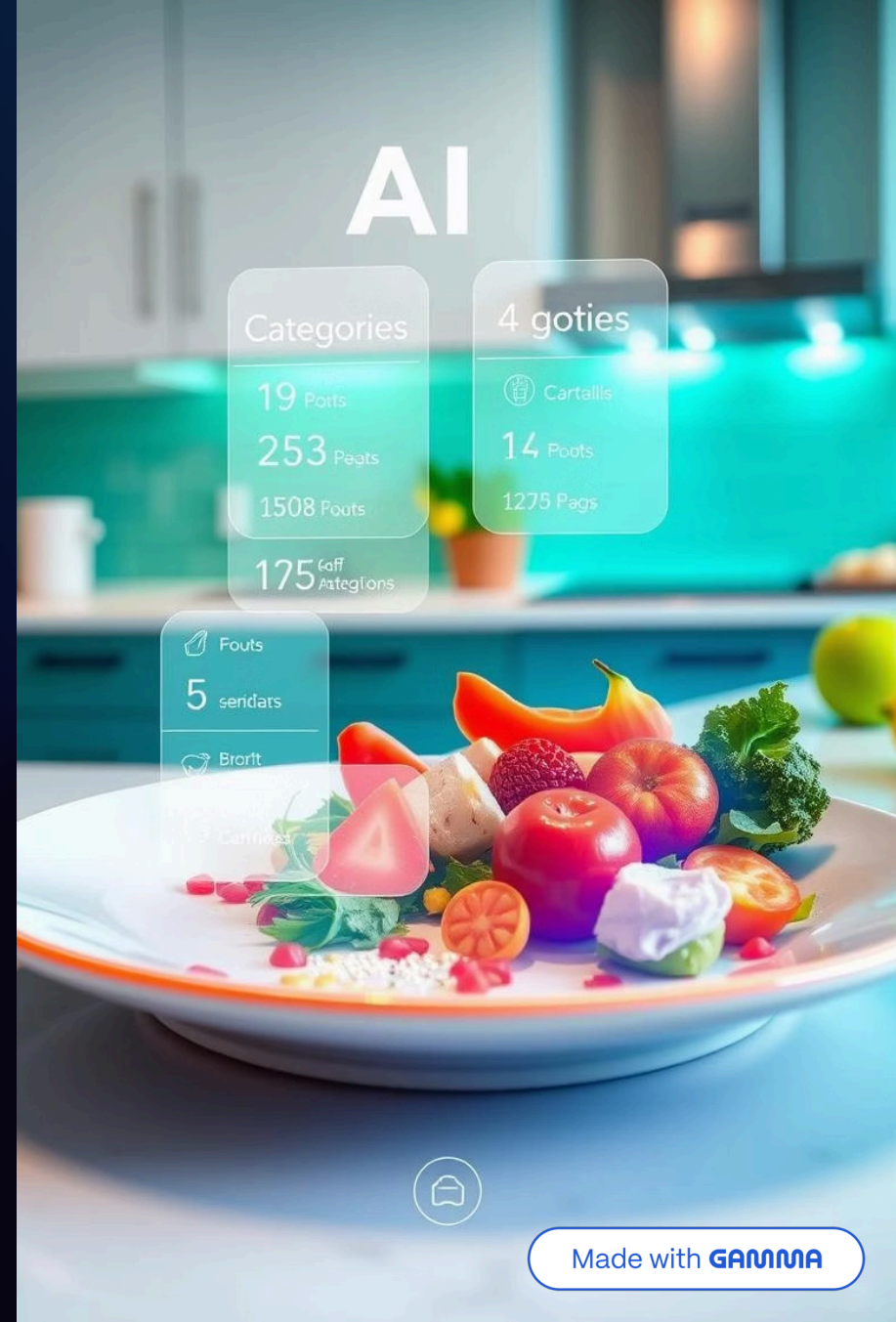


So3ratk: Real-time Food Detection & Classification

Discover how our innovative system leverages advanced AI to instantly identify and categorise food items, revolutionising various applications from dietary tracking to inventory management.



Meet the Team

Our project's success is driven by a dedicated team with diverse expertise, each playing a crucial role in bringing So3ratk to life.



Mohamed Ahmed Mohamed

YOLO Pretrained Model Retrieval



Mohamed Abdelaziem Sayed

Backend Development



Hussien Elhaddad

Frontend Development



Hossam Ahmed

Web Deployment Specialist



Mario Saber

Model Deployment & Integration



Mohamed Shehab

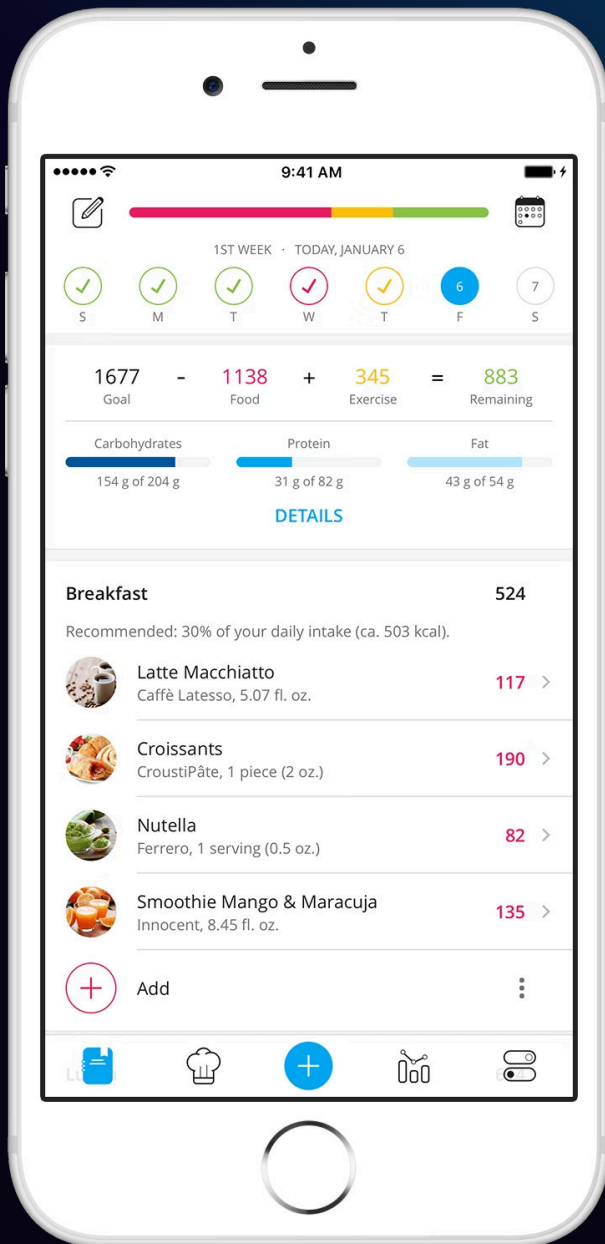
Testing & Dataset Retrieval

The Challenge: Dietary Tracking for Egyptian Cuisine

Egyptians face significant hurdles in managing their diets effectively due to a critical gap in available nutritional resources. Global food tracking applications consistently lack accurate data for traditional Egyptian dishes.

- ❏ This absence of specific macro-nutrient information makes precise dietary planning and calorie counting nearly impossible, hindering healthy eating efforts.





1. Introduction

So3ratk solves this with an automated system that identifies and classifies food directly from images. Powered by YOLO, it delivers fast and accurate detection, built with enough efficiency and scalability to work reliably in real-world scenarios.

So3ratk – Food Detection

So3ratk is an end-to-end deep learning system designed for real-time food detection and classification using advanced computer vision and YOLO models. It lets users upload food images and instantly receive accurate results with bounding boxes and labels.



Accuracy

Utilizes YOLO model for better accuracy of detecting and reconginzing food



Speed

providing immediate insights without delay.



Real-time Processing

Engineered for real-time performance, offering dynamic analysis as images are uploaded.

2. Project Objectives



Detection System

Build a high-accuracy food detection pipeline using modern deep learning models.



Dataset

Select and preprocess a curated Egyptian food dataset for optimal model training.



Model Training & Optimization

Train and fine-tune a YOLO model for maximum precision and real-time inference speed..



Web System Integration

Connect the trained model to a full frontend and backend interface for seamless user interaction.



Real-time Application Deployment

deploy a fully accessible, real-time food detection application, making it available for public use.

3. System Architecture Flow

1

Dataset

Collection and preprocessing of a diverse food image dataset.

2

YOLO Model

Core deep learning model for fast, accurate food detection.

3

RESTful API

Handle inference requests and return detection results.

4

Web Frontend

Upload images and display predictions in real time.

5

Deployment

Deployed using **Streamlit** for real-time demonstration.

6

Testing

Test the deployed model and detect any issues

4. Dataset Collection (Milestone 1)

4.1 Data Collection

The foundational dataset for So3ratk with a rich collection of Egyptian food images acquired

[egypt-food Object Detection Dataset by meto](#)





4.2 Data Annotation

All images in the dataset were carefully annotated, ensuring each food item is accurately labeled to support effective training and evaluation of the detection model.

4.3 Exploratory Data Analysis



EDA Insights

Exploratory Data Analysis provided critical insights into class distribution, resolution consistency, and training sample balance per class.



Dataset Splitting

The dataset was rigorously partitioned into distinct training, validation, and testing sets to ensure robust model evaluation.

5. Model Development and Training (Milestone 2)

Model Selection

YOLO architecture chosen for its high speed and accuracy, initialized with pre-trained weights for transfer learning.

Environment Setup

Development environment included Python, PyTorch, OpenCV, and the Ultralytics YOLO framework.

Model Training

Trained the dataset over multiple epochs, employing transfer learning for accelerated convergence.

Model Evaluation

Performance assessed using precision, recall, and analysis of training/validation loss curves.

Predicted Food Classes

Our model is trained to accurately detect and classify the following nine traditional Egyptian dishes, offering precise identification for dietary tracking and more.



Bechamel



Molokhia



Ataif



Besala



Fool



Konafa



Koshary



Basposa



Taamia

6. Backend System Integration (Milestone 3)



API Development

A robust RESTful backend API was developed using Flask, ensuring efficient and scalable communication.



Image Uploads

The API is designed to seamlessly accept image uploads from various client applications.



YOLO Inference

Upon image reception, the API executes the YOLO model for swift and accurate food detection inference.



JSON Responses

Results are returned as structured JSON responses, detailing detected food classes, bounding box coordinates, and confidence scores.



Output Format

The API response adheres to a standardized format, facilitating seamless integration with the frontend for a cohesive user experience.

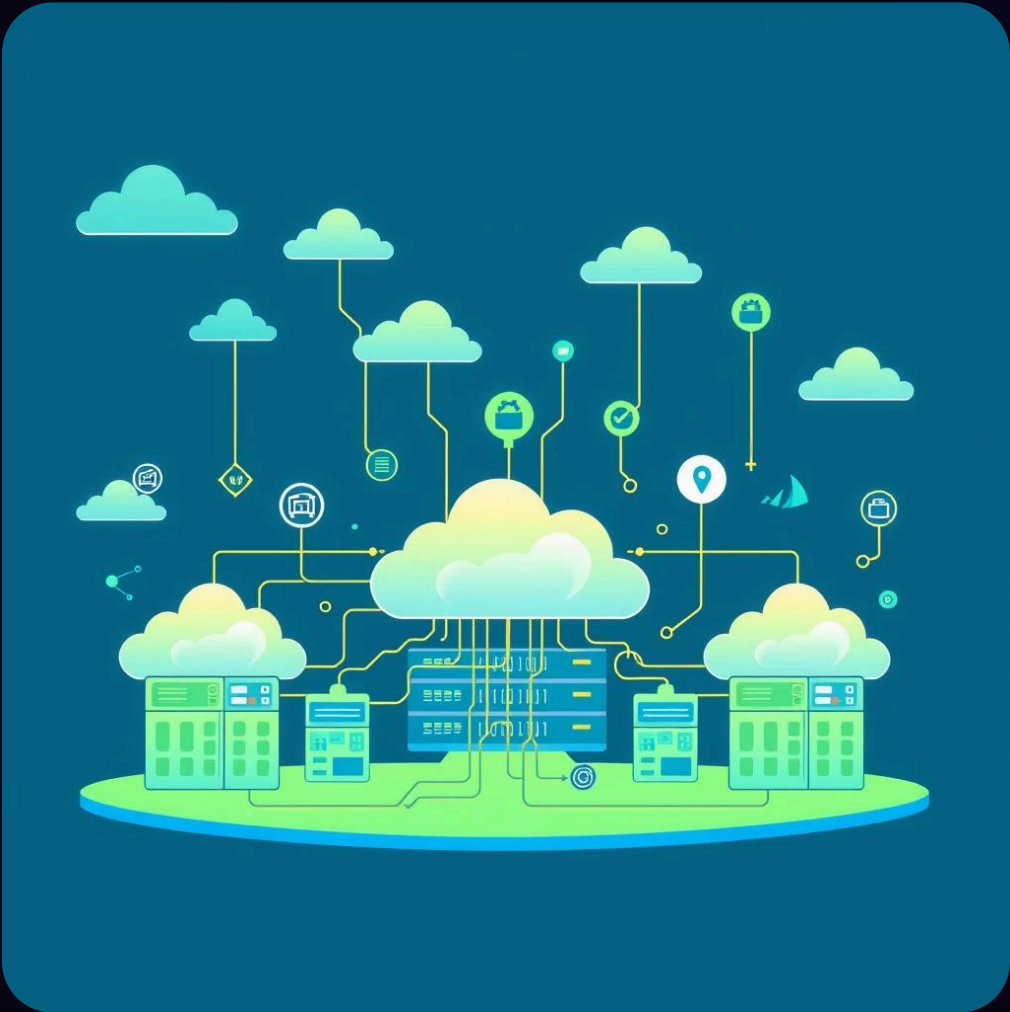
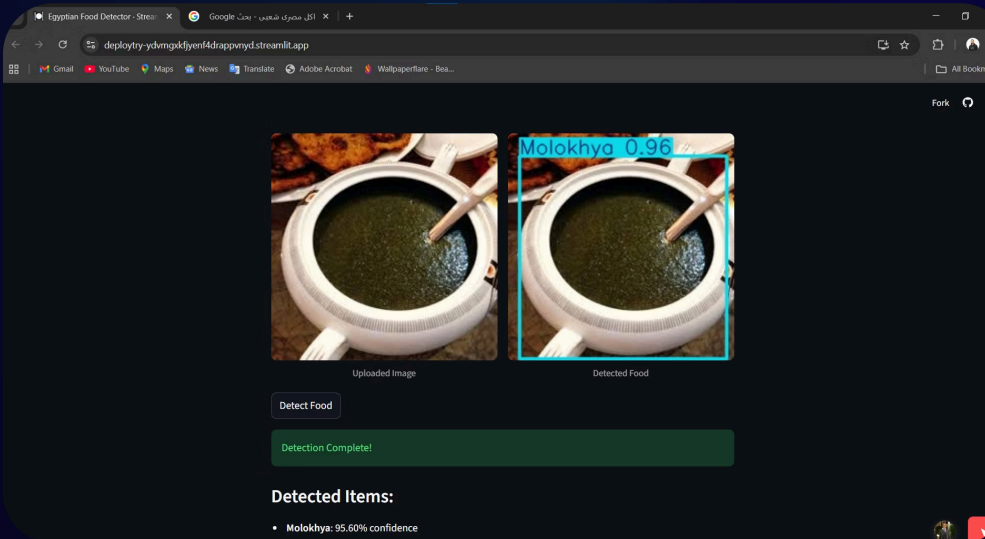
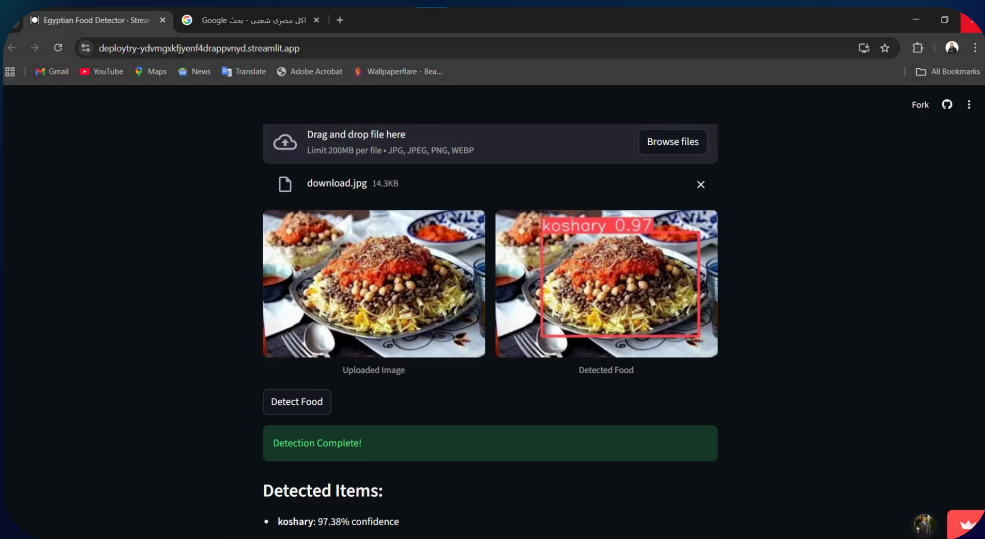
7. Web Interface and Deployment (Milestone 4)

7.1 Frontend Development

A clean and intuitive web application was crafted using a combination of HTML, CSS, and JavaScript. This interface empowers users to effortlessly upload food images and instantly visualize the detection results. The focus was on creating a highly responsive and user-friendly experience.

7.2 Visualization

Detected food items are presented with clear visual cues: bounding boxes are precisely drawn directly onto the uploaded image, accompanied by distinct food class labels and their corresponding confidence percentages. This ensures immediate comprehension of the model's output.



7.3 API Integration

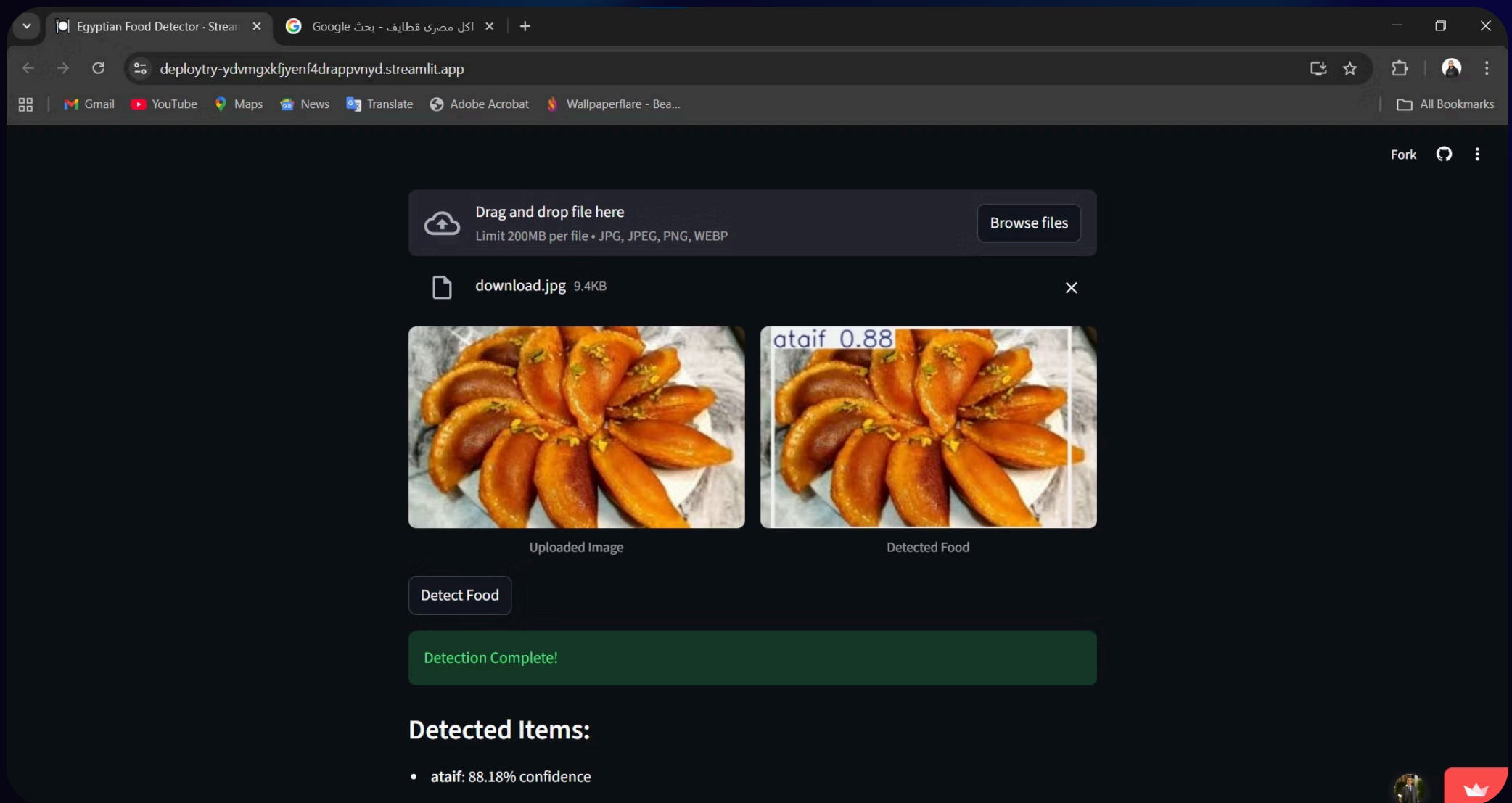
The frontend establishes robust communication with Flask backend API, efficiently sending user-uploaded images for processing and asynchronously retrieving the rich detection results. This client-server architecture ensures a smooth and interactive user journey.

7.4 Deployment

The entire So3ratk system was strategically deployed using Streamlit cloud platform. This cloud-native approach guarantees high availability, scalability, and performance, enabling public access to the real-time food detection service from anywhere in the world.

So3ratk Live Demo

Get ready to experience So3ratk firsthand. We will now navigate to our web application to showcase its real-time food detection and classification capabilities. (<https://deploytry-ydvmgxfjyenf4drappvnyd.streamlit.app/>)



Future Work

To further enhance So3ratk's utility and accuracy, our future development will focus on integrating advanced nutritional data and pioneering AR-driven estimation capabilities.



Nutritional Databases

Integrate extensive databases for calories and macronutrients (proteins, fats, carbs) to provide detailed dietary insights for each detected food item.



AR Volume Estimation

Develop and integrate Augmented Reality (AR) features to accurately estimate the volume of detected food portions directly through the user's camera.



Precise Calorie Calculation

Establish a robust mechanism to convert estimated food volumes into precise weight measurements, subsequently calculating accurate calorie counts and macro breakdowns.

Thank You

We appreciate your time and attention. Please feel free to ask any questions you may have about So3ratk, our process, or future plans.

